

ALTAI, Mongolia

WMO: 442770

Lat: 46.3775N

Lon: 96.2378E

Elev: 7156

StdP: 11.27

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-20.6	-31.9	1.2	-21.0	-27.2	1.6	-17.3	41.0	7.8	35.7	11.0	2.8	270	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	77.1	56.4	74.1	54.9	71.5	54.2	62.4	72.4	59.9	69.5	57.6	67.0	10.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.7	96.6	68.6	56.0	87.4	65.6	53.6	79.9	63.7	32.0	72.6	29.9	69.8	28.1	67.1	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
35.9	29.1	24.3	DB	-29.7	81.9	4.9	2.5	-33.2	83.7	-36.1	85.2	-38.9	86.6	-42.5	88.4	
			WB	-29.4	62.9	4.9	6.4	-32.9	67.5	-35.8	71.2	-38.6	74.8	-42.1	79.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.2	-0.2	5.4	18.2	34.4	45.0	55.8	60.3	56.8	45.6	31.1	15.4	4.9
	DBStd	22.87	10.60	10.67	11.45	9.69	8.53	6.41	5.79	6.25	8.02	9.49	10.58	10.03
	HDD50	7692	1557	1249	984	472	198	21	2	15	173	587	1037	1397
	HDD65	12370	2022	1669	1449	919	620	282	167	261	581	1051	1487	1862
	CDD50	831	0	0	0	3	44	194	322	226	42	0	0	0
	CDD65	35	0	0	0	0	0	5	22	8	0	0	0	0
	CDH74	270	0	0	0	0	1	48	169	52	0	0	0	0
	CDH80	22	0	0	0	0	0	3	17	2	0	0	0	0
Wind	WSAvg	8.8	6.7	7.1	9.6	11.8	11.3	8.8	7.9	8.0	9.2	9.2	8.8	7.3
Precipitation	PrecAvg	6.7	0.0	0.1	0.2	0.4	0.6	1.1	1.6	1.5	0.6	0.3	0.1	0.1
	PrecMax	12.1	0.2	0.4	0.6	1.2	1.9	4.1	4.6	4.0	3.9	1.3	0.4	0.3
	PrecMin	3.7	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0
	PrecStd	2.1	0.0	0.1	0.2	0.2	0.4	0.7	1.1	1.0	0.6	0.3	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	34.4	48.9	62.9	72.0	79.4	82.4	78.6	70.1	57.5	44.9	32.2
		MCWB	25.2	28.0	36.4	47.5	52.7	55.1	61.9	56.7	50.1	42.1	35.4	25.7
	2%	DB	25.0	29.3	43.8	57.4	67.8	74.6	78.4	75.0	65.8	52.6	40.0	27.9
		MCWB	20.5	23.8	32.7	41.8	49.9	54.4	58.1	54.4	48.6	39.9	32.1	23.1
	5%	DB	21.2	26.2	39.9	53.6	63.9	71.9	75.3	72.0	62.4	49.1	35.3	24.2
		MCWB	17.2	21.4	30.2	39.4	47.0	53.8	56.2	54.1	47.2	37.8	29.0	20.2
	10%	DB	17.1	22.6	35.8	49.8	60.2	68.9	72.3	69.1	59.1	45.9	31.3	20.5
		MCWB	14.2	18.5	27.8	37.4	44.9	52.0	55.2	53.2	45.6	35.9	26.2	17.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	28.7	39.2	49.3	57.3	62.0	67.7	64.0	55.6	46.2	36.5	27.0
		MCDB	29.0	33.1	46.3	60.3	70.0	74.6	78.9	73.6	66.1	55.7	42.5	31.0
	2%	WB	20.9	24.5	34.0	44.4	52.8	59.2	63.9	60.8	52.3	41.9	32.6	23.5
		MCDB	24.5	28.6	41.8	53.2	64.5	70.6	74.5	69.7	61.4	49.5	39.1	27.5
	5%	WB	17.3	21.5	30.5	41.0	49.5	56.8	61.2	58.4	49.5	39.3	29.4	20.4
		MCDB	20.7	25.8	38.6	50.4	60.0	67.7	70.9	67.4	57.9	46.9	34.9	24.1
	10%	WB	14.1	18.6	28.1	38.2	46.6	54.6	58.8	56.1	46.9	36.7	26.4	17.1
		MCDB	17.3	22.6	35.6	48.2	56.9	64.4	67.4	64.3	56.3	44.7	30.9	20.3
Mean Daily Temperature Range	5% DB	MDBR	21.3	22.2	21.4	19.0	19.4	19.0	17.6	18.2	19.0	18.9	19.1	19.8
		MCDBR	22.3	23.6	22.7	22.3	23.2	23.2	21.3	21.7	21.2	21.7	21.0	21.8
		MCWBR	18.8	19.3	15.5	13.3	13.3	12.3	10.5	10.9	11.6	13.8	15.4	18.1
	5% WB	MCDBR	22.3	22.9	21.9	21.2	21.9	20.6	19.1	19.6	19.7	20.9	20.6	21.3
		MCWBR	19.6	19.3	16.3	15.0	15.0	13.6	12.5	12.8	13.3	15.1	16.3	18.1
Clear-Sky Solar Irradiance	taub	0.230	0.261	0.305	0.346	0.350	0.360	0.356	0.333	0.299	0.280	0.250	0.233	
	taud	2.369	2.287	2.204	2.173	2.225	2.278	2.331	2.378	2.432	2.402	2.385	2.355	
	Ebn at Noon	285	292	292	288	290	286	286	290	292	282	272	269	
	Edh at Noon	22	30	38	43	43	41	38	35	30	26	22	20	
All-Sky Solar Radiation	RadAvg	605	963	1445	1899	2230	2231	2136	1903	1615	1098	659	493	
	RadStd	44	42	70	76	130	116	102	80	60	58	56	31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-5.90	N/A	-4.07	-4.34	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	-3.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page